Google Summer of Code 2026

Machine Learning for Science (ML4SCI)

Brain-to-Brain Decoder: Validating Neural Synchrony Patterns in Human Conversation

Sourish Kumar Dutta

B.Tech | IIT (ISM) Dhanbad, India

duttasourish12@gmail.com | github.com/sourishkumardutta

March 2026

Project	NEURODYAD — Brain-to-Brain Decoder
Organization	ML4SCI / University of Alabama
Mentors	Dr. Evie Malaia (Univ. of Alabama) & Dr. Brendan Ames (Univ. of Southampton)
Duration	350 hours, 12 weeks (Full-time)
Difficulty	Intermediate
Contact	ml4-sci@cern.ch

Abstract

This proposal outlines my plan to contribute to the NeuroDyads project for GSoC 2026 under ML4SCI. The project aims to build a rigorous, explainable machine-learning pipeline for hyperscanning EEG data recorded from conversing dyads, extending the pilot work from GSoC 2025. My contributions will focus on three core deliverables: (1) a stratified cross-validation pipeline for dyadic EEG datasets, (2) cross-entropy distance metrics for comparing CEBRA embedding distributions across neurotype groups, and (3) explainability analyses mapping embedding features to brain regions and frequency bands. Building directly on my pre-task submission — a complete end-to-end preprocessing and CEBRA embedding pipeline with 98.35% KNN decoding accuracy — I am technically prepared to make immediate, meaningful contributions on day one.

About Me

I am a first-year B.Tech student in Petroleum Engineering at IIT (ISM) Dhanbad, pursuing computer science and machine learning as a self-driven parallel discipline. My interest lies at the intersection of representation learning, neural signal analysis, and AI for scientific discovery — fields I have been building depth in entirely outside my formal curriculum.

My technical foundation spans PyTorch, NumPy, scikit-learn, MNE-Python, and CEBRA. I have built neural networks from scratch — including a scalar autograd engine validated against PyTorch — competed on Kaggle, and earned certifications in Machine Learning (Stanford/Coursera) and CNNs (DeepLearning.AI). I am an active member of CyberLabs, IIT ISM's premier ML research club.

What draws me to the NeuroDyads project specifically is the problem of interpersonal brain dynamics. The prospect of learning whether two brains genuinely coordinate during conversation — and which neural signatures drive that coordination — is a question I find both scientifically compelling and socially important, given its implications for autism research and assistive technology.

Why NeuroDyads and ML4SCI

Most ML research optimizes within a single domain. NeuroDyads sits at the boundary of contrastive representation learning, clinical neuroscience, and social cognition — and asks genuinely open questions. The 2025 pilot (Glushanina et al., arXiv 2025) demonstrated that CEBRA can decode speaker-listener roles and demographic features from joint EEG with high accuracy. But the mechanism remains a black box: which brain regions matter? Which frequency bands? How do embedding structures compare across dyads with different neurotypes?

These are the questions that the 2026 project is designed to answer, and they are exactly the kind of rigorous, explainability-focused problems I want to work on. The Crompton et al. (2025, Nature Human Behaviour) finding — that communication breakdowns stem from neurotype mismatch rather than individual deficits — gives this work direct clinical relevance.

I also chose ML4SCI because the organization has a demonstrated track record of producing research-grade outputs through GSoC. Last year's NeuroDyads preprint is a direct testament to that. I want to be part of work that reaches arXiv — not just a summer project.

Pre-Task: What I Built and What I Learned

My pre-task submission implements the complete NeuroDyads pipeline from raw EDF files to interpreted CEBRA embeddings. Below is a summary of the key technical choices I made and why they matter for the 2026 project.

Part 1 — Preprocessing

I loaded two raw 65-channel EDF files (Speaker: NT#10, Listener: NT#9) at 250 Hz, verified zero-offset DIN1 markers, applied a 0.1–45 Hz bandpass filter (Hamming FIR), dropped the VREF reference channel, renamed channels to GSN-HydroCel-65_1.0 standard labels, and set the electrode montage for ICA compatibility.

Bad channel detection via PyPrep (RANSAC): 10 bad channels in the Speaker file (E7, E21, E23, E27, E29, E31, E32, E52, E62, E63) and 6 in the Listener (E10, E23, E41, E55, E62, E63) were identified by deviation, HF noise, correlation, and RANSAC criteria and interpolated using spherical splines. This step is essential because CEBRA assumes the joint T×128 matrix is meaningful at every time step — a noisy or flat channel poisons every sample in which it participates.

ICA artifact removal: I selected Extended Infomax ICA for both participants, with 20 components for the Speaker and 15 for the Listener. The component count was justified using the data-to-component rule ($30N^2$ data points required): 20 components use ~16% of the Speaker's 75,411 samples, squarely in the reliable range. I excluded 8 components from the Speaker (eyeblinks: ICA003, ICA004, ICA008; saccades: ICA006; jaw/muscle: ICA001, ICA009, ICA015; electrode artifact: ICA019) and 4 from the Listener (eye movements: ICA001, ICA002, ICA004; cardiac/eye: ICA006). Crucially, I chose to perform ICA separately per participant — joining two different brains into a single ICA decomposition would violate the statistical independence assumption and cause artifactual cross-participant mixing. I documented this decision explicitly in the notebook.

One observation I noted: after ICA, the back-projected signal showed slightly elevated high-frequency power due to rank deficiency from bad channel interpolation. This is a known mathematical artifact of ICA reconstruction on rank-reduced data, not a signal quality issue. I documented the reasoning in the notebook.

Improvement over the 2025 baseline: The original NeuroDyads pipeline used 15-component ICA with a fixed exclusion array. In reviewing the topographies, I identified additional ocular artifacts (ICA002, ICA004 — distinct vertical blinks) and a saccade component (ICA006) that were not excluded in the baseline. I removed these additional components from the Speaker data to reduce CEBRA's tendency to learn from large-variance eye movements rather than true neural dynamics.

Part 2 — CEBRA Embedding

I concatenated the preprocessed Speaker and Listener data into a T×128 matrix (first 64 columns = Speaker, last 64 = Listener), z-normalized each channel independently across time, and trained CEBRA with affective labels (0 = positive, 1 = negative) using a time-based 80/20 train-test split (preserving temporal structure to prevent leakage).

CEBRA configuration: offset10-model, cosine distance, time_delta conditioning, 3D output, 5000 iterations, batch size 512, learning rate 3e-4. This closely follows the 2025 pipeline (Glushanina et al.) with time_offsets=5 instead of 10 to match the shorter segment length of this single-dyad dataset.

Results — Main model: KNN-5 decoding accuracy

KNN-5 decoding accuracy: 98.35% | InfoNCE Loss (GoF): 5.74

Control model (shuffled labels): 49.45% | GoF: 6.17

The control model's accuracy collapsed to near-chance (49.45%), confirming that the main model's structure is label-driven and not a consequence of temporal autocorrelation alone.

Part 3 — Embedding Geometry & Part 4 — Critical Reflection

In Part 3, I observed two well-separated 3D clusters (positive vs. negative affect) with a sparse transition zone, interpretable as the neural boundary between conversational states. Temporal coloring revealed sub-clusters within each condition, suggesting moment-to-moment modulations in joint brain state.

Part 4 (the most important section) identified the core limitation: the binary affect label (one flip over the entire recording) gives CEBRA almost no within-condition variability to learn from. The model likely exploits temporal position as a proxy for the label, since both co-vary perfectly. My time_delta conditioning compounds this. A proper cross-shift control (offsetting one participant's stream by ~30 seconds) would disentangle whether CEBRA learned dyadic co-occurrence or just individual temporal trajectories — I identified this missing control as the single most important thing I would add with more time.

I also identified that independent ICA per participant breaks the synchronized artifact rejection that the 2025 paper used (removing identical time segments from both streams simultaneously). This is a preprocessing alignment issue I plan to address systematically in the GSoC pipeline.

Project Plan

The 2026 NeuroDyads project scales from 8 to 40+ dyads and shifts focus from proof-of-concept decoding to rigorous validation and explainability. My 12-week plan is organized around three deliverables, each building on the last.

Deliverable 1 — Stratified Cross-Validation Pipeline (Weeks 1–4)

The central methodological challenge with dyadic EEG data is that standard cross-validation leaks information: if trials from the same dyad appear in both train and test sets, the model can overfit to pair-specific signatures rather than learning generalizable patterns. I will build a dyad-aware stratified cross-validation framework that treats each dyad as an indivisible unit.

- Implement leave-one-dyad-out (LODO) CV as the primary evaluation scheme, consistent with the N+1 analysis in Glushanina et al. (2025)
- Implement stratified k-fold (k=5) at the dyad level, ensuring proportional representation of neurotype combinations (autistic-autistic, autistic-neurotypical, neurotypical-neurotypical) in each fold
- Implement the cross-shift control: offset Participant B's time series by a fixed lag (~30 s) and retrain CEBRA to test whether dyadic co-occurrence — not individual temporal trajectories — drives decoding accuracy
- Generate learning curves: KNN accuracy and GoF as a function of number of dyads, to determine the sample size at which neural decoding stabilises and justify the 40+ dyad target
- Reproduce the pseudocut sanity check from the 2025 paper on the full dataset to confirm that preprocessing cuts do not introduce embedding artifacts

Deliverable 2 — Cross-Entropy Distance Metrics (Weeks 5–8)

The Roca et al. (2023, Cell Reports Methods) cross-entropy test provides a principled, model-free way to compare embedding distributions across groups. Rather than comparing summary statistics (mean accuracy, GoF), cross-entropy distance tells us whether the geometric structure of two embeddings is the same — that is, whether the same neural dynamics underlie neural synchrony in same-neurotype vs. mixed-neurotype dyads.

- Implement the cross-entropy distance pipeline from Roca et al. 2023: train a reference CEBRA model on one group, compute the encoding of a second group's data, and measure the distributional distance between embeddings
- Apply to three group comparisons: (a) same-neurotype (autistic-autistic vs. neurotypical-neurotypical), (b) mixed vs. unmixed dyads, (c) same-gender vs. cross-gender dyads (replicating the gender-related finding from the 2025 N+1 analysis)
- Compute confidence intervals via bootstrap resampling and assess statistical significance
- Visualize the distance matrix across dyad groups and annotate with AQ score differences to assess whether neurotype mismatch continuously predicts embedding distance

Deliverable 3 — Explainability: Feature-to-Brain Mapping (Weeks 9–11)

The 2025 ablation analysis (removing channels associated with speech production or executive function) found no significant impact on decoding accuracy, suggesting distributed neural representations. I will go further: rather than removing channels and checking whether performance degrades, I will identify which channels and frequency bands positively contribute to the learned embedding.

- Implement channel contribution analysis: for each of the 128 channels (64 per participant), compute the change in InfoNCE loss when that channel's time series is replaced with Gaussian noise. Rank channels by contribution magnitude and visualize as a topographic map
- Implement frequency band decomposition: bandpass-filter the raw EEG into standard bands (delta 1–4 Hz, theta 4–8 Hz, alpha 8–12 Hz, beta 12–30 Hz, gamma 30–45 Hz) and train separate CEBRA models per band to determine which frequencies drive decoding
- Combine channel and frequency analyses: identify which (electrode, frequency) pairs most inform classification, and compare spatial maps between speaker and listener to test for hemispheric or regional asymmetries in inter-brain synchrony
- Relate the top contributing features to known neurophysiology (e.g., theta-band prefrontal synchrony in social cognition, alpha-band temporal regions in speech comprehension)

Buffer / Documentation Week (Week 12)

- Finalize all code, write comprehensive docstrings, and integrate the pipeline into the NeuroDyads GitHub repository
- Prepare a brief technical write-up summarizing methods and key findings, suitable for inclusion in the project's forthcoming publication
- Address any feedback from mentors, resolve open issues, and ensure reproducibility on the full 40+ dyad dataset

Timeline at a Glance

Phase	Dates	Deliverables
Community Bonding (May)	May 8 – June 1	Deep-read Roca et al. 2023 and full NeuroDyads codebase. Set up environment. Discuss prioritization with mentors.
Weeks 1–2	June 2–15	Dyad-aware LODO and stratified k-fold CV framework. Learning curve generation.
Weeks 3–4	June 16–29	Cross-shift control. Pseudocut sanity check on full dataset. Preliminary cross-dyad results.
Weeks 5–6	June 30–July 13	Cross-entropy distance pipeline (Roca et al.). Group comparison: neurotype and gender.
Weeks 7–8	July 14–27	Bootstrap confidence intervals. Distance matrix visualization. Midterm evaluation.
Weeks 9–10	July 28–Aug 10	Channel contribution (noise masking) analysis. Topographic maps.
Week 11	Aug 11–17	Frequency band decomposition. Combined (channel, band) explainability maps.
Week 12	Aug 18–25	Integration, documentation, write-up, mentor review.

Technical Skills & Relevant Experience

Directly Relevant

- MNE-Python: Full preprocessing pipeline (filtering, bad channel detection, ICA, segmentation) demonstrated in the NeuroDyads pre-task
- CEBRA: Trained and evaluated contrastive embedding models, including control analyses and GoF metrics, on real hyperscanning EEG data
- PyPrep: RANSAC-based automated bad channel detection
- NumPy, SciPy, scikit-learn: Data manipulation, z-normalization, KNN classification, train-test splits
- PyTorch: Built autograd engines and neural networks from scratch (Micrograd project, validated against PyTorch autograd)

Supporting

- ML competition experience: Finished 3rd on Kaggle private leaderboard (CyberLabs WOC); demonstrated OOF cross-validation, class imbalance handling, and probability calibration
- 2nd place, AI Summit 2026 (among 60–80 teams) — autonomous job-search agent using FastAPI and Gemini API
- Certifications: ML Specialization (Stanford/Coursera, Dec 2025), CNNs (DeepLearning.AI, Mar 2026)
- Active member, CyberLabs ML Division, IIT (ISM) Dhanbad (Jan 2026 – present)

Commitment and Availability

GSoC 2026 will be my sole major commitment during the coding period. I have no summer internship, coursework conflicts, or competing obligations. I can commit 35–40 hours per week throughout the 12-week period.

I am in IST (UTC+5:30). I am comfortable communicating asynchronously via the project mailing list (ml4-sci@cern.ch) and synchronously via video call at times convenient for mentors in the US and UK. I will share weekly progress updates and push code to GitHub on a regular cadence.

I have already engaged deeply with the project: I have read the NeuroDyads preprint (arXiv 2509.23479v1), the CEBRA paper (Schneider et al., Nature 2023), the Roca et al. cross-entropy paper (Cell Reports Methods 2023), and the Crompton et al. communication paper (Nature Human Behaviour 2025) referenced in the project description. I reviewed the original NeuroDyads GitHub repository and used it to contextualize my preprocessing decisions in the pre-task.

Conclusion

The NeuroDyads project addresses a genuinely hard problem: understanding how two brains coordinate during natural conversation, and what that coordination tells us about neurotypical and neurodivergent social communication. The 2025 GSoC pilot proved feasibility. The 2026 project asks the harder question of why it works — which features matter, how robust the results are, and how embedding structures compare across groups.

I have done more than read about this project. I have built the pipeline, made my own data-driven decisions at every step, identified its real limitations, and documented my reasoning in detail. My pre-task submission — 98.35% KNN accuracy on a genuinely cleaned dyadic EEG dataset, with a rigorous control analysis and a specific, honest critical reflection — is evidence that I can make meaningful research contributions, not just run tutorials.

I would be honoured to work with Dr. Malaia and Dr. Ames to push this project forward.